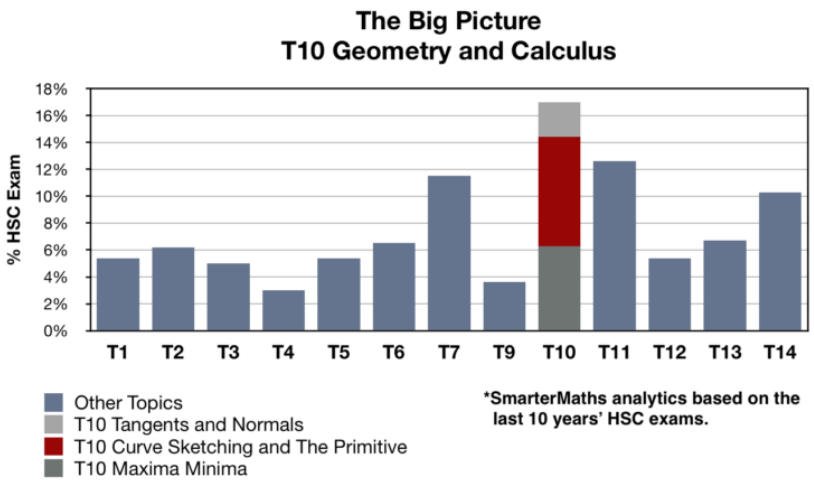
ADV: Calculus (Adv), C1 Introduction to Differentiation (Adv)

Standard Differentiation (Y11)
Tangents (Y11)
Rates of Change (Y11)

Teacher: Jason Hu

Exam Equivalent Time: 99 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*



HISTORICAL CONTRIBUTION

- *T10 Geometry and Calculus* is the single largest topic within the Mathematics course, contributing 17.0% to each paper, on average, in the last 10 years.
- This topic has been split into three sub-categories: *1-Maxima and Minima (6.3%), 2-Curve Sketching and The Primitive Function (8.1%), and 3-Tangents and Normals (2.6%).*

HSC ANALYSIS - What to expect and common pitfalls

- *Tangents and Normals (2.6%)* questions make up the smallest sub-topic within *Topic 10*. It is consistently tested and offers up very achievable marks - fitting within our definition of low lying fruit.
- This topic area has appeared in the HSC in 9 of the last 10 years (including 2018).
- Examiners have increased the difficulty level of tangent questions in the past by including them within longer answer cross-topic problems. Note that *The Parabola* and *Logs and Exponentials* are the most common cross-topic partners in these harder questions.

Questions

1. Calculus, 2ADV C1 2016 HSC 11b

   RAP Data - Bottom 9%: School result (98%) was 8% above state average (90%)

   Differentiate $\dfrac{x+2}{3x-4}$. *(2 marks)*

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2. Calculus, 2ADV C1 2009 HSC 1d

   RAP Data - Bottom 20%: School result (98%) was 13% above state average (85%)

   Find the gradient of the tangent to the curve $y = x^4 - 3x$ at the point $(1, -2)$. *(2 marks)*

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3. Calculus, 2ADV C1 2012 HSC 11c

   Find the equation of the tangent to the curve $y = x^2$ at the point where $x = 3$. *(2 marks)*

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4. Calculus, 2ADV C1 SM-Bank 1

   i. Use differentiation by first principles to find y', given $y = 2x^2 + 5x$. *(2 marks)*

   ii. Find the equation of the tangent to the curve when $x = 1$. *(1 mark)*

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5. Calculus, 2ADV C1 2018 HSC 12d

   The displacement of a particle moving along the x-axis is given by

   $$x = \frac{t^3}{3} - 2t^2 + 3t,$$

   where x is the displacement from the origin in metres and t is the time in seconds, for $t \geq 0$.

   (i) What is the initial velocity of the particle? *(1 mark)*

   (ii) At which times is the particle stationary? *(2 marks)*

   (iii) Find the position of the particle when the acceleration is zero. *(2 marks)*

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6. Calculus, 2ADV C1 2004 HSC 1b

   Differentiate $x^4 + 5x^{-1}$ with respect to x. *(2 marks)*

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7. Calculus, 2ADV C1 2005 HSC 2bii

Differentiate with respect to x:

$$\frac{x^2}{x-1}.$$ *(2 marks)*

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8. Calculus, 2ADV C1 2014 HSC 11c

Differentiate $\dfrac{x^3}{x+1}$. *(2 marks)*

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9. Calculus, 2ADV C1 2015 HSC 12c

RAP Data - Bottom 15%: School result (95%) was 11% above state average (84%)

Find $f'(x)$, where $f(x) = \dfrac{x^2+3}{x-1}$. *(2 marks)*

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10. Calculus, 2ADV C1 2019 HSC 11c

RAP Data - Bottom 18%: School result (98%) was 12% above state average (86%)

Differentiate $\dfrac{2x+1}{x+5}$. *(2 marks)*

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11. Calculus, 2ADV C1 2011 HSC 2c

RAP Data - Bottom 22%: School result (91%) was 13% above state average (78%)

Find the equation of the tangent to the curve $y = (2x+1)^4$ at the point where $x = -1$. *(3 marks)*

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12. Calculus, 2ADV C1 2017 HSC 12a

Find the equation of the tangent to the curve $y = x^2 + 4x - 7$ at the point $(1, -2)$. *(2 marks)*

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13. Calculus, 2ADV C1 SM-Bank 2

i. Find the equations of the tangents to the curve $y = x^2 - 3x$ at the points where the curve cuts the x-axis. *(2 marks)*

ii. Where do the tangents intersect? *(2 marks)*

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14. Calculus, 2ADV C1 SM-Bank 3

The displacement x metres from the origin at time t seconds of a particle travelling in a straight line is given by

$$x = 2t^3 - t^2 - 3t + 11 \quad \text{when } t \geq 0$$

i. Calculate the velocity when $t = 2$. *(1 mark)*

ii. When is the particle stationary? *(2 marks)*

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15. Calculus, 2ADV C1 2013 HSC 11b

Evaluate $\displaystyle\lim_{x \to 2} \frac{(x-2)(x+2)^2}{x^2-4}$. *(2 marks)*

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16. Calculus, 2ADV C1 2014 HSC 13c

The displacement of a particle moving along the x-axis is given by

$$x = t - \frac{1}{1+t},$$

where x is the displacement from the origin in metres, t is the time in seconds, and $t \geq 0$.

(i) Show that the acceleration of the particle is always negative. *(2 marks)*

(ii) What value does the velocity approach as t increases indefinitely? *(1 mark)*

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17. Calculus, 2ADV C1 SM-Bank 25

Let $g(x) = \left(2 - x^3\right)^3$.

Evaluate $g'(-1)$. *(2 marks)*

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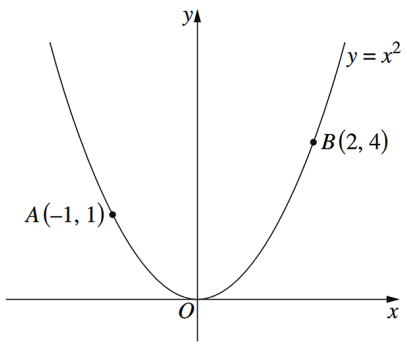
18. Calculus, 2ADV C1 2010 HSC 7b

The parabola shown in the diagram is the graph $y = x^2$. The points $A(-1, 1)$ and $B(2, 4)$ are on the parabola.



(i) Find the equation of the tangent to the parabola at A. ***(2 marks)***

(ii) Let M be the midpoint of AB.

There is a point C on the parabola such that the tangent at C is parallel to AB.

Show that the line MC is vertical. ***(2 marks)***

(iii) The tangent at A meets the line MC at T.

Show that the line BT is a tangent to the parabola. ***(2 marks)***

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19. Calculus, 2ADV C1 2006 HSC 8a

A particle is moving in a straight line. Its displacement, x metres, from the origin, O, at time t seconds, where $t \geq 0$, is given by $x = 1 - \dfrac{7}{t+4}$.

(i) Find the initial displacement of the particle. ***(1 mark)***

(ii) Find the velocity of the particle as it passes through the origin. ***(3 marks)***

(iii) Show that the acceleration of the particle is always negative. ***(1 mark)***

(iv) Sketch the graph of the displacement of the particle as a function of time. ***(2 marks)***

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20. Calculus, 2ADV C1 2019 HSC 14d

The equation of the tangent to the curve $y = x^3 + ax^2 + bx + 4$ at the point where $x = 2$ is $y = x - 4$.

Find the values of a and b. ***(3 marks)***

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21. Calculus, 2ADV C3 2010 HSC 8d

RAP Data - Bottom 18%: School result (40%) was 12% above state average (28%)

Let $f(x) = x^3 - 3x^2 + kx + 8$, where k is a constant.

Find the values of k for which $f(x)$ is an increasing function. ***(2 marks)***

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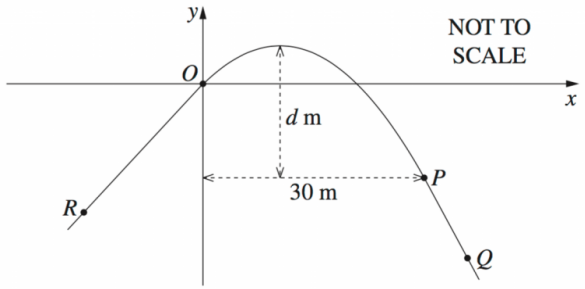
22. Calculus, 2ADV C1 2009 HSC 6c

The diagram illustrates the design for part of a roller-coaster track. The section RO is a straight line with slope 1.2, and the section PQ is a straight line with slope -1.8. The section OP is a parabola $y = ax^2 + bx$. The horizontal distance from the y-axis to P is 30 m.



In order that the ride is smooth, the straight line sections must be tangent to the parabola at O and at P.

(i) Find the values of a and b so that the ride is smooth. ***(3 marks)***

(ii) Find the distance d, from the vertex of the parabola to the horizontal line through P, as shown on the diagram. ***(2 marks)***

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Worked Solutions

1. Calculus, 2ADV C1 2016 HSC 11b

$$y = \frac{x+2}{3x-4}$$

Using the quotient rule:

$$\left(\frac{g}{h}\right)' = \frac{g'h - gh'}{h^2}$$

$$y' = \frac{1(3x-4) - (x+2)\cdot 3}{(3x-4)^2}$$

$$= \frac{-10}{(3x-4)^2}$$

2. Calculus, 2ADV C1 2009 HSC 1d

$$y = x^4 - 3x$$

$$\frac{dy}{dx} = 4x^3 - 3$$

At $x = 1$

$$\frac{dy}{dx} = 4 - 3 = 1$$

$\therefore$ Gradient of tangent at $(1, -2) = 1$.

Worked Solutions

3. Calculus, 2ADV C1 2012 HSC 11c

$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

Need to find equation with m = 6, through (3,9)

When $x = 3, y = 9, \frac{dy}{dx} = 6$

Using $y - y_1 = m(x - x_1)$

$$y - 9 = 6(x - 3)$$

$$y - 9 = 6x - 18$$

$$6x - y - 9 = 0$$

$\therefore$ Equation of the tangent is 6x - y - 9 = 0

4. Calculus, 2ADV C1 SM-Bank 1

i. $f(x) = 2x^2 + 5x$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\left(2(x+h)^2 + 5(x+h)\right) - \left(2x^2 + 5x\right)}{h}$$

$$= \lim_{h \to 0} \frac{2x^2 + 4xh + h^2 + 5x + 5h - 2x^2 - 5x}{h}$$

$$= \lim_{h \to 0} \frac{4xh + h^2 + 5h}{h}$$

$$= \lim_{h \to 0} \frac{h(4x + 5 + h)}{h}$$

$$\therefore\ y' = 4x + 5$$

ii. When $x = 1, y = 7$

$y' = 4 + 5 = 9$

$$\therefore y - 7 = 9(x - 1)$$

$$y = 9x - 2$$

5. Calculus, 2ADV C1 2018 HSC 12d

(i) $x = \dfrac{t^3}{3} - 2t^2 + 3t$

$$v = \frac{dx}{dt} = t^2 - 4t + 3$$

Find v when $t = 0$:

$v = 0 - 0 + 3$

$= 3 \text{ ms}^{-1}$

(ii) Particle is stationary when $v = 0$

$t^2 - 4t + 3 = 0$

$(t - 3)(t - 1) = 0$

$t = 1$ or 3 seconds

(iii) $a = \dfrac{dv}{dt} = 2t - 4$

Find t when $a = 0$

$2t - 4 = 0$

$t = 2$

$$x(2) = \frac{2^3}{3} - 2\left(2^2\right) + 3(2)$$

$$= \frac{8}{3} - 8 + 6$$

$$= \frac{2}{3}$$

6. Calculus, 2ADV C1 2004 HSC 1b

$$y = x^4 + 5x^{-1}$$

$$\frac{dy}{dx} = 4x^3 - 5x^{-2}$$

7. Calculus, 2ADV C1 2005 HSC 2bii

$$y = \frac{x^2}{x-1}$$

Using $\dfrac{dy}{dx} = \dfrac{u'v - uv'}{v^2}$

$$u = x^2 \qquad v = x - 1$$
$$u' = 2x \qquad v' = 1$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{2x(x-1) - x^2(1)}{(x-1)^2} \\ &= \frac{2x^2 - 2x - x^2}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2}\end{aligned}$$

8. Calculus, 2ADV C1 2014 HSC 11c

$$y = \frac{x^3}{x+1}$$

Using $\dfrac{dy}{dx} = \dfrac{u'v - uv'}{v^2}$

$$u = x^3 \qquad v = (x+1)$$
$$u' = 3x^2 \qquad v' = 1$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3x^2(x+1) - x^3(1)}{(x+1)^2} \\ &= \frac{3x^3 + 3x^2 - x^3}{(x+1)^2} \\ &= \frac{2x^3 + 3x^2}{(x+1)^2} \\ &= \frac{x^2(2x+3)}{(x+1)^2}\end{aligned}$$

9. Calculus, 2ADV C1 2015 HSC 12c

$$f(x) = \frac{x^2+3}{x-1}$$

Using the quotient rule

$$u = x^2 + 3 \qquad v = x - 1$$
$$u' = 2x \qquad v' = 1$$

$$\begin{aligned}f'(x) &= \frac{u'v - uv'}{v^2} \\ &= \frac{2x(x-1) - \left(x^2+3\right) \times 1}{(x-1)^2} \\ &= \frac{2x^2 - 2x - x^2 - 3}{(x-1)^2} \\ &= \frac{x^2 - 2x - 3}{(x-1)^2} \\ &= \frac{(x-3)(x+1)}{(x-1)^2}\end{aligned}$$

10. Calculus, 2ADV C1 2019 HSC 11c

Using quotient rule:

$$u = 2x + 1, \quad v = x + 5$$
$$u' = 2, \qquad v' = 1$$

$$\begin{aligned}y' &= \frac{u'v - v'u}{v^2} \\ &= \frac{2(x+5) - (2x+1)}{(x+5)^2} \\ &= \frac{2x + 10 - 2x - 1}{(x+5)^2} \\ &= \frac{9}{(x+5)^2}\end{aligned}$$

11. Calculus, 2ADV C1 2011 HSC 2c

$y = (2x+1)^4$

Using the Function of a Function Rule (or Chain Rule)

$$\frac{dy}{dx} = 4 \times (2x+1)^3 \times \frac{d}{dx}(2x+1)$$
$$= 8(2x+1)^3$$

At $x = -1,\ y = 1$

$$\frac{dy}{dx} = 8(2(-1)+1)^3$$
$$= 8(-1)^3$$
$$= -8$$

MARKER'S COMMENT: The best setting out clearly showed the derivative function, the gradient, the point and then finally, calculations for the equation of the tangent, as per the Worked Solution.

Tangent has $m = -8$ through $(-1, 1)$

Using $y - y_1 = m(x - x_1)$

$$y - 1 = -8(x+1)$$
$$y - 1 = -8x - 8$$
$$8x + y + 7 = 0$$

$\therefore$ Equation of tangent is $8x + y + 7 = 0$

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12. Calculus, 2ADV C1 2017 HSC 12a

$y = x^2 + 4x - 7$

$$\frac{dy}{dx} = 2x + 4$$

When $x = 1$, $\frac{dy}{dx} = 6$

Equation of tangent through $(1, -2)$

$$y + 2 = 6(x-1)$$
$$y = 6x - 8$$

---

13. Calculus, 2ADV C1 SM-Bank 2

i. $y = x^2 - 3x$

$$= x(x-3)$$

Cuts x-axis at $x = 0$ or $x = 3$

$$\frac{dy}{dx} = 2x - 3$$

At $x = 0 \Rightarrow \frac{dy}{dx} = -3$

T_1 has $m = -3$, through $(0, 0)$

$$y - 0 = -3(x - 0)$$
$$y = -3x$$

At $x = 3 \Rightarrow \frac{dy}{dx} = 3$

T_2 has $m = 3$, through $(3, 0)$

$$y - 0 = 3(x - 3)$$
$$y = 3x - 9$$

ii. Intersection occurs when:

$$3x - 9 = -3x$$
$$6x = 9$$
$$x = \frac{3}{2}$$

$$y = -3 \times \frac{3}{2} = -\frac{9}{2}$$

$\therefore$ Intersection at $\left(\frac{3}{2}, -\frac{9}{2}\right)$

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14. Calculus, 2ADV C1 SM-Bank 3

i. $x = 2t^3 - t^2 - 3t + 11$

$$v = \frac{dx}{dt} = 6t^2 - 2t - 3$$

When $t = 2$,

$$v = 6 \times 2^2 - 2 \cdot 2 - 3$$
$$= 17 \text{ ms}^{-1}$$

ii. Particle is stationary when $v = 0$

$$6t^2 - 2t - 3 = 0$$

$$\therefore t = \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 6 \cdot (-3)}}{12}$$
$$= \frac{2 \pm \sqrt{76}}{12}$$
$$= \frac{1 \pm \sqrt{19}}{6}$$
$$= \frac{1 + \sqrt{19}}{6} \quad (t \geq 0)$$

15. Calculus, 2ADV C1 2013 HSC 11b

$$\lim_{x \to 2} \frac{(x-2)(x+2)^2}{x^2 - 4}$$
$$= \lim_{x \to 2} \frac{(x-2)(x+2)^2}{(x-2)(x+2)}$$
$$= \lim_{x \to 2} (x+2)$$
$$= 4$$

COMMENT: This question has been simplified as students no longer need to factorise the difference between 2 cubes ($x^3 - 2^3$).

16. Calculus, 2ADV C1 2014 HSC 13c

(i) $x = t - \dfrac{1}{1+t}$

$$= t - (1+t)^{-1}$$

$$\dot{x} = 1 - (-1)(1+t)^{-2}$$
$$= 1 + \frac{1}{(1+t)^2}$$

$$\ddot{x} = -2(1+t)^{-3}$$
$$= -\frac{2}{(1+t)^3}$$

Since $t \geq 0$,

$$\Rightarrow -\frac{2}{(1+t)^3} < 0$$

$\therefore$ Acceleration is always negative.

(ii) Velocity $(\dot{x}) = 1 + \dfrac{1}{(1+t)^2}$

As $t \to \infty$, $\dfrac{1}{(1+t)^2} \to 0$

$\therefore$ As $t \to \infty$, $\dot{x} \to 1$

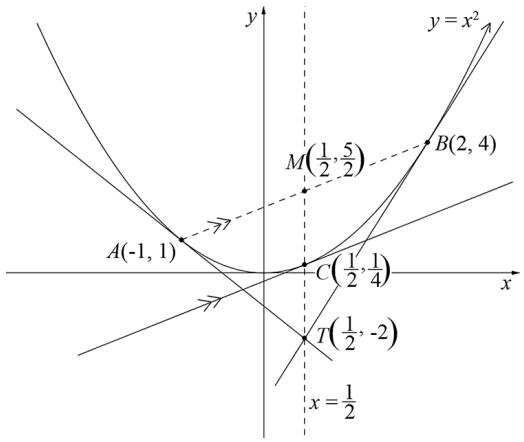
17. Calculus, 2ADV C1 SM-Bank 25

Using Chain Rule:

$$g'(x) = 3(2 - x^3)^2(-3x^2)$$
$$= -9x^2(2 - x^3)^2$$
$$\therefore g'(1) = -9(-1)^2\left[2 - (-1)^3\right]^2$$
$$= -81$$

18. Calculus, 2ADV C1 2010 HSC 7b

(i)



$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

At $A(-1,1) \Rightarrow \dfrac{dy}{dx} = -2$

Tangent has $m = -2$, through $(-1,1)$:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -2(x + 1)$$

$$y - 1 = -2x - 2$$

$$2x + y + 1 = 0$$

$\therefore$ Tangent at A is $2x + y + 1 = 0$

(ii) $A(-1,1)$ $B(2,4)$

$$M = \left(\frac{-1+2}{2}, \frac{1+4}{2}\right)$$

$$= \left(\frac{1}{2}, \frac{5}{2}\right)$$

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

♦ *Mean mark 37%.*
IMPORTANT: The critical understanding required for this question is that the gradient of AB needs to be equated to the gradient function (i.e. $\dfrac{dy}{dx}$).

$$= \frac{4-1}{2+1} = 1$$

When $\frac{dy}{dx} = 1$

$$2x = 1$$

$$x = \frac{1}{2}$$

$\therefore\ C\left(\frac{1}{2}, \frac{1}{4}\right)$

$\Rightarrow M$ and C both have x-value $= \frac{1}{2}$

$\therefore\ MC$ is vertical ... as required

(iii) T is point on tangent when $x = \frac{1}{2}$

Tangent $2x + y + 1 = 0$

At $x = \frac{1}{2}$

$$2 \times \left(\frac{1}{2}\right) + y + 1 = 0$$

$$\Rightarrow y = -2$$

$\therefore\ T\left(\frac{1}{2}, -2\right)$

Given $B(2, 4)$

$$m_{BT} = \frac{4+2}{2 - \frac{1}{2}}$$

$$= 4$$

At $B(2, 4)$, find gradient of tangent

$$\frac{dy}{dx} = 2x = 2 \times 2 = 4$$

$\therefore m_{\text{tangent}} = 4 = m_{BT}$

$\therefore\ BT$ is a tangent

♦♦ *Mean mark 29%.*

19. Calculus, 2ADV C1 2006 HSC 8a

(i) $x = 1 - \frac{7}{t+4}$

When $t = 0$,

$$x = 1 - \frac{7}{4}$$

$$= -\frac{3}{4} \text{ m}$$

$\therefore$ Initial displacement is $\frac{3}{4}$ metres to the left of the origin.

(ii) $x = 1 - \frac{7}{t+4} = 1 - 7(t+4)^{-1}$

$$\dot{x} = (-1) - 7(t+4)^{-2} \times \frac{d}{dt}(t+4)$$

$$= 7(t+4)^{-2} \times 1$$

$$= \frac{7}{(t+4)^2}$$

Find t when $x = 0$

$$0 = 1 - \frac{7}{t+4}$$

$$\frac{7}{t+4} = 1$$

$$7 = (t+4)$$

$$t = 3$$

When $t = 3$

$$\dot{x} = \frac{7}{(3+4)^2}$$

$$= \frac{1}{7} \text{ ms}^{-1}$$

$\therefore$ The velocity of the particle as it passes

through the origin is $\frac{1}{7}$ ms^{-1}.

(iii) $\dot{x} = 7(t+4)^{-2}$

$$\ddot{x} = \frac{d\dot{x}}{dt} = -14\,(t+4)^{-3}$$

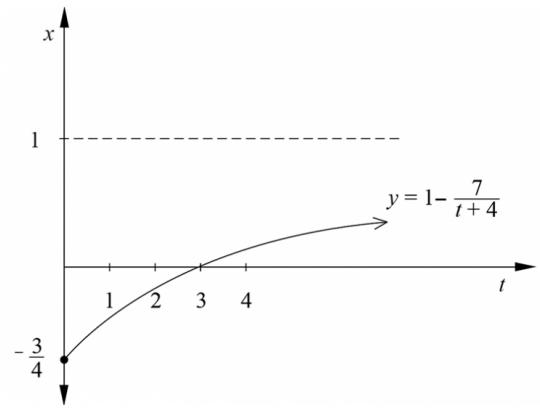
Given $t \geq 0$

$$\Rightarrow\ (t+4)^{-3} \geq 0$$

$$\Rightarrow\ -14(t+4)^{-3} \leq 0$$

$\therefore\ \ddot{x}$ is always negative.

(iv)



20. Calculus, 2ADV C1 2019 HSC 14d

$$y = x^3 + ax^2 + bx + 4$$

$$\frac{dy}{dx} = 3x^2 + 2ax + b$$

When $x = 2$, $\frac{dy}{dx} = 1$

$$12 + 4a + b = 1$$

$$4a + b = -11 \ldots (1)$$

♦ Mean mark 46%.

The point $(2, -2)$ lies on y:

$$8 + 4a + 2b + 4 = -2$$

$$4a + 2b = -14 \ldots (2)$$

Subtract $(2) - (1)$

$$b = -3$$

Substitute into (1)

$$4a - 3 = -11$$

$$4a = -8$$

$$a = -2$$

21. Calculus, 2ADV C3 2010 HSC 8d

$$f(x) = x^3 - 3x^2 + kx + 8$$
$$f'(x) = 3x^2 - 6x + k$$

$f(x)$ is increasing when $f'(x) > 0$

$\Rightarrow 3x^2 - 6x + k > 0$

$f'(x)$ is always positive

$\Rightarrow f'(x)$ is a positive definite.

i.e. when $a > 0$ and $\Delta < 0$

$$a = 3 > 0$$
$$\Delta = b^2 - 4ac$$
$$\therefore (-6)^2 - (4 \times 3 \times k) < 0$$
$$36 - 12k < 0$$
$$12k > 36$$
$$k > 3$$

$\therefore$ $f(x)$ is increasing when $k > 3$.

♦♦ *Mean mark 28%.*
MARKER'S COMMENT: The arithmetic required to solve $36 - 12k < 0$ proved the undoing of *too many* students in this question. *TAKE CARE!*

22. Calculus, 2ADV C1 2009 HSC 6c

(i) $y = ax^2 + bx$

$$\frac{dy}{dx} = 2ax + b$$

At $x = 0$,

$$\frac{dy}{dx} = b$$

We need m at $O = 1.2$

$\therefore$ $b = 1.2$

At P, $x = 30$

$$\frac{dy}{dx} = 2 \times a \times 30 + 1.2$$
$$= 60a + 1.2$$

We need m at $P = -1.8$

$$60a + 1.2 = -1.8$$
$$60a = -3$$
$$a = -\frac{3}{60}$$
$$= -0.05$$

$\therefore$ For a smooth ride, $a = -0.05$ and $b = 1.2$

♦♦♦ *Exact data unavailable.*
MARKER'S COMMENT: Successful students equated the curve gradient to the straight section as a requirement for a smooth ride.

(ii) $y = -0.05x^2 + 1.2x$

$$\frac{dy}{dx} = -0.1x + 1.2$$

Find x when $\frac{dy}{dx} = 0$

$$-0.1x + 1.2 = 0$$
$$x = \frac{1.2}{0.1}$$
$$= 12$$

$\therefore$ MAX when $x = 12$

When $x = 12$

$$\begin{aligned} y &= -0.05 \times 12^2 + 1.2 \times 12 \\ &= -7.2 + 14.4 \\ &= 7.2 \end{aligned}$$

When $x = 30$

$$\begin{aligned} y &= -0.05 \times 30^2 + 1.2 \times 30 \\ &= -45 + 36 \\ &= -9 \end{aligned}$$

$$\begin{aligned} \therefore\ d &= 7.2 + |-9| \\ &= 16.2\ \text{m} \end{aligned}$$

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